

BOULOGNE, France

WMO: 070020

Lat: 50.7325N

Lon: 1.5997E

Elev: 243

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.3	28.6	17.1	13.1	28.9	20.7	15.7	30.6	39.2	48.2	34.5	47.4	9.8	70	0.651

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	79.4	66.0	75.3	64.3	71.9	62.7	67.5	76.5	65.6	72.6	64.0	69.6	8.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	90.3	70.3	62.8	86.2	68.2	61.6	82.8	66.8	32.0	77.0	30.5	72.4	29.3	70.0	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.0	28.2	25.4	DB	23.0	87.1	4.8	4.7	19.6	90.5	16.8	93.2	14.1	95.8	10.6	99.2	(4)
(5)			WB	21.2	71.5	4.5	1.7	17.9	72.7	15.2	73.7	12.7	74.6	9.4	75.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	41.2	41.6	45.0	49.6	54.5	59.3	63.0	64.1	60.5	55.1	48.0	42.9
	DBStd	9.66	6.37	5.76	5.19	5.58	5.47	4.74	4.52	4.23	3.96	5.00	5.22	6.40
	HDD50	1101	275	239	167	71	13	0	0	0	0	15	95	226
	HDD65	4799	739	656	619	463	329	184	95	66	144	309	511	684
	CDD50	1880	1	2	13	59	154	280	403	438	316	172	35	7
	CDD65	104	0	0	0	1	5	14	33	39	11	1	0	0
	CDH74	482	0	0	0	1	19	79	168	185	29	1	0	0
	CDH80	102	0	0	0	0	2	17	40	40	3	0	0	0
Wind	WSAvg	12.4	14.4	13.6	12.6	11.0	11.3	10.6	11.2	11.2	11.5	13.0	13.5	14.5
Precipitation	PrecAvg	33.7	3.1	2.3	1.9	2.0	2.3	2.2	2.3	2.6	2.8	3.8	4.4	4.0
	PrecMax	45.6	5.7	6.1	6.0	5.8	4.8	5.2	5.3	5.3	6.3	9.5	10.4	8.1
	PrecMin	22.5	0.4	0.6	0.2	0.1	0.2	0.9	0.3	0.4	0.1	0.9	1.3	0.9
	PrecStd	6.0	1.4	1.4	1.2	1.4	1.1	1.1	1.2	1.5	1.5	2.0	2.2	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	56.8	62.9	71.1	76.5	82.1	84.6	85.3	77.8	71.3	60.9	55.6
		MCWB	50.3	48.4	52.0	56.9	61.8	67.0	67.5	68.7	65.2	61.8	56.9	52.6
	2%	DB	51.5	52.1	57.8	65.8	71.7	75.2	78.4	79.2	72.2	66.0	58.1	54.0
		MCWB	49.5	47.4	50.6	54.6	60.9	64.1	65.3	66.3	63.1	60.2	55.3	51.7
	5%	DB	50.3	50.1	54.3	61.7	67.2	70.7	73.9	74.6	68.5	63.2	56.6	52.6
		MCWB	48.2	47.4	48.9	52.7	58.0	61.9	64.0	64.8	61.3	59.0	53.9	50.5
	10%	DB	48.9	48.5	51.4	57.6	63.2	66.8	69.9	70.5	65.9	61.3	55.2	51.2
		MCWB	46.8	46.4	47.3	50.9	56.4	59.7	62.4	62.9	60.2	57.8	52.4	49.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	50.3	54.0	58.2	65.1	68.2	70.0	70.6	66.8	63.0	58.2	53.8
		MCDB	52.3	52.7	59.2	67.4	72.8	79.3	82.2	81.7	73.8	67.7	59.8	54.7
	2%	WB	49.9	49.0	51.5	55.7	61.4	65.3	67.0	67.6	64.6	61.2	56.1	52.2
		MCDB	51.2	50.8	56.2	64.0	68.9	73.7	75.8	75.9	69.9	64.8	57.7	53.5
	5%	WB	48.5	47.8	49.6	53.6	58.9	62.6	64.8	65.6	62.9	59.7	54.5	50.9
		MCDB	49.8	49.5	53.0	59.9	65.3	69.4	71.5	72.3	66.5	62.6	56.3	52.4
	10%	WB	47.2	46.6	48.2	51.6	56.7	60.5	63.0	64.0	61.6	58.4	52.7	49.3
		MCDB	48.7	48.4	50.6	56.6	61.9	65.7	68.3	69.1	64.8	61.0	54.9	50.9
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.5	7.8	9.5	9.3	9.4	8.8	8.7	8.6	7.8	6.6	6.4
		MCDBR	6.6	7.8	12.6	15.2	15.9	16.8	16.6	15.7	12.8	9.6	7.2	6.0
		MCWBR	6.7	6.0	8.1	8.6	9.2	9.7	8.5	7.7	7.1	6.6	6.6	6.1
	5% WB	MCDBR	6.1	6.4	10.3	13.7	14.4	15.1	14.2	12.9	9.9	8.6	6.5	5.9
		MCWBR	6.6	5.8	7.4	8.2	9.1	9.5	8.1	7.2	6.6	7.0	6.8	6.4
Clear-Sky Solar Irradiance	taub	0.329	0.346	0.383	0.402	0.410	0.410	0.407	0.400	0.379	0.364	0.338	0.322	
	taud	2.394	2.385	2.284	2.255	2.288	2.312	2.324	2.367	2.408	2.431	2.417	2.380	
	Ebn at Noon	220	247	257	266	269	269	268	264	257	238	214	202	
	Edh at Noon	20	26	34	39	40	39	38	35	30	25	20	18	
All-Sky Solar Radiation	RadAvg	263	488	871	1401	1662	1840	1763	1474	1097	621	304	202	
	RadStd	22	71	105	120	117	152	143	103	69	57	33	24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page